Engineering Economy (INDE 301)

Chapters 7 & 8 Homework — Solution Key

Rate of Return Analysis: Single & Multiple Alternatives

Instructor: Naim Al Khoury · Teaching Assistant: Mohammad Younis

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Part A — Single-Alternative Rate of Return

Exercise 1: Machinery Upgrade for a Manufacturing Facility

Solution:

Given: First cost \$400,000; uniform savings \$75,000 per year for 8 years; MARR = 12%. Later additions: salvage value \$60,000 at Year 8; mid-life overhaul \$80,000 at Year 5.
Find: i^* at each stage, unrecovered balance schedule, and economic acceptability.

Part 1. Complete Present Worth equation defining i^* in standard factor notation:

$$PW(i^*) = -400{,}000 + 75{,}000\,(P/A,\, i^*,\, 8) = 0$$

Part 2. *Rate of return i^* for the base project:* Because the savings are a pure uniform series, setting the present worth factor equal to the cost-to-benefit ratio yields:

$$(P/A,\, i_0,\, 8) = \frac{\$400{,}000}{\$75{,}000} = 5.3333$$

From interest factor tables for $n = 8$, $(P/A, 10\%, 8) = 5.3349$. Therefore, $i_0 \approx 10\%$ is an immediate starting estimate. Evaluating PW at 10% and 12% (2% apart):

$$\begin{aligned} PW(10\%) &= -400{,}000 + 75{,}000(5.3349) \\ &= \$117.50 > 0 \\ PW(12\%) &= -400{,}000 + 75{,}000(4.9676) \\ &= -\$27{,}430.00 < 0 \end{aligned}$$

Linear interpolation between 10% and 12%:

$$\begin{aligned} i^* &\approx 10\% + \frac{\$117.50}{\$117.50 - (-\$27{,}430.00)}\,(2\%) \\ &= 10\% + \frac{117.50}{27{,}547.50}\,(2\%) \\ &= 10.01\% \end{aligned}$$

(Exact Excel value: `=RATE(8, 75000, -400000)` = 10.01%.)

Part 3. Since $i^* \approx 10.01\% < \text{MARR} = 12\%$ (confirmed by $PW(12\%) = -\$27{,}430.00 < 0$), the base investment is **not acceptable**.

Part 4. *Unrecovered Balance Schedule at $i^* = 10.01\%$:*

Year	Beginning Balance	Interest Charged (10.01%)	Cash Flow	Recovered Principal	Ending Balance
0	\$0.00	\$0.00	−\$400,000.00	—	−\$400,000.00
1	−\$400,000.00	−\$40,040.00	\$75,000.00	\$34,960.00	−\$365,040.00
2	−\$365,040.00	−\$36,540.50	\$75,000.00	\$38,459.50	−\$326,580.50

Step-by-Step Derivation:

$$\text{Year 1 Interest} = -\$400{,}000.00 \times 10.01\% = -\$40{,}040.00$$
$$\text{Year 1 Ending Balance} = -\$400{,}000.00 - \$40{,}040.00 + \$75{,}000.00 = -\$365{,}040.00$$
$$\text{Year 2 Interest} = -\$365{,}040.00 \times 10.01\% = -\$36{,}540.50$$
$$\text{Year 2 Ending Balance} = -\$365{,}040.00 - \$36{,}540.50 + \$75{,}000.00 = -\$326{,}580.50$$

Critical Takeaway: Interest in Year 2 is charged strictly on the unrecovered balance (−\$365,040.00), dropping to −\$36,540.50. If simple flat interest were charged on the initial principal, the interest charge would remain −\$40,040.00 every single year. This confirms the fundamental definition taught in Course Notes Section 7.1.

Part 5. Updated Present Worth equation including the Year 8 salvage value:

$$PW(i^*) = -400{,}000 + 75{,}000\,(P/A,\, i^*,\, 8) + 60{,}000\,(P/F,\, i^*,\, 8) = 0$$

Part 6. *Initial Estimate using the Course Notes Manual Procedure and Linear Interpolation:* Following Course Notes Section 7.2.1, the cash flows consist of a uniform annual series of \$75,000 for 8 years and a terminal salvage value of \$60,000 at Year 8. Because the vast majority of receipts are uniform ($8 \times \$75{,}000 = \$600{,}000$ vs. \$60,000), we convert the single salvage value into an equivalent annual receipt without considering the time value of money:

$$A' = \$75{,}000 + \frac{\$60{,}000}{8} = \$75{,}000 + \$7{,}500 = \$82{,}500$$
$$(P/A,\, i_0,\, 8) = \frac{\$400{,}000}{\$82{,}500} = 4.8485$$

From the interest tables for $n = 8$, $(P/A,\, 12\%,\, 8) = 4.9676$ and $(P/A,\, 14\%,\, 8) = 4.6389$, yielding an initial estimate of $i_0 \approx 12.72\%$.[1]

[1] If a student alternatively converted all receipts into a single future sum at Year 8 without TVM

Following the Course Notes refinement procedure, we evaluate PW near i_0:

- **Trial 1 at $i = 12\%$:**

$$\begin{aligned} PW(12\%) &= -400{,}000 + 75{,}000\,(P/A,\,12\%,\,8) + 60{,}000\,(P/F,\,12\%,\,8) \\ &= -400{,}000 + 75{,}000(4.9676) + 60{,}000(0.4039) \\ &= -\$3{,}196.00 < 0 \end{aligned}$$

  Since $PW(12\%) < 0$, the return is less than 12%.

- **Trial 2 at $i = 10\%$:**

$$\begin{aligned} PW(10\%) &= -400{,}000 + 75{,}000\,(P/A,\,10\%,\,8) + 60{,}000\,(P/F,\,10\%,\,8) \\ &= -400{,}000 + 75{,}000(5.3349) + 60{,}000(0.4665) \\ &= \$28{,}107.50 > 0 \end{aligned}$$

Since PW changes sign between 10% and 12%, we linearly interpolate:

$$\begin{aligned} i^* &\approx 10\% + \frac{\$28{,}107.50}{\$28{,}107.50 - (-\$3{,}196.00)}\,(2\%) \\ &= 10\% + \frac{28{,}107.50}{31{,}303.50}\,(2\%) \\ &= 11.80\% \end{aligned}$$

(Exact Excel value: `=IRR` $= 11.78\%$.)

Part 7. *Baseline Anchoring Shortcut:* Because adding a positive salvage value adds $\$60{,}000(P/F, i, 8) > 0$ to the Present Worth equation, the Present Worth at any given interest rate is strictly greater than the base Present Worth. Using the base trial rate $i = 10\%$:

$$\begin{aligned} PW(10\%) &= PW_{\text{base}}(10\%) + 60{,}000\,(P/F,\,10\%,\,8) \\ &= \$117.50 + 60{,}000(0.4665) \\ &= \$28{,}107.50 > 0 \end{aligned}$$

Because $PW(10\%) > 0$, the updated root must exceed 10%. Testing the next standard

($F = 8(\$75{,}000) + \$60{,}000 = \$660{,}000$), $(P/F, i_0, 8) = 400{,}000/660{,}000 = 0.6061 \Rightarrow i_0 \approx 6.46\%$. This deviates significantly from the true root because it ignores compounding on the uniform series, illustrating the importance of choosing the predominant factor as taught in Course Notes Section 7.2.1.

table rate, $i = 12\%$:

$$\begin{aligned} PW(12\%) &= PW_{\text{base}}(12\%) + 60{,}000\,(P/F,\, 12\%,\, 8) \\ &= -\$27{,}430.00 + 60{,}000(0.4039) \\ &= -\$3{,}196.00 < 0 \end{aligned}$$

The root is instantly bracketed between 10% and 12% without requiring an initial estimate i_0 from scratch. This illustrates how previously computed rates can be leveraged to streamline trial-and-error root finding.

Part 8. The \$60,000 salvage value raises the rate of return from 10.01% to 11.78% (an increase of 1.77%). However, because 11.78% < MARR = 12% (check: $PW(12\%) = -\$3{,}196.00 < 0$), the project remains **not acceptable**.

Part 9. Updated Present Worth equation including the mid-life overhaul of \$80,000 at $t = 5$:

$$PW(i^*) = -400{,}000 + 75{,}000\,(P/A,\, i^*,\, 8) - 80{,}000\,(P/F,\, i^*,\, 5) + 60{,}000\,(P/F,\, i^*,\, 8) = 0$$

Part 10. *Full Calculation of Initial Estimate i_0 from Scratch and Linear Interpolation:* Following Course Notes Section 7.2.1, we do not start from previous results and instead determine the initial estimate directly from the complete cash flows:

1. **Convert disbursements into a single initial amount without TVM:**

$$PW_D = \$400{,}000 + \$80{,}000 = \$480{,}000$$

2. **Convert receipts into a uniform series without TVM:** Because over 90% of receipts form an 8-year uniform series ($8 \times \$75{,}000 = \$600{,}000$ vs. \$60,000 salvage), we convert the single salvage value into an equivalent annual receipt:

$$A' = \$75{,}000 + \frac{\$60{,}000}{8} = \$75{,}000 + \$7{,}500 = \$82{,}500$$

3. **Solve $PW_D = PW_R$ for the initial guess factor:**

$$\begin{aligned} \$480{,}000 &= \$82{,}500\,(P/A,\, i_0,\, 8) \\ (P/A,\, i_0,\, 8) &= \frac{\$480{,}000}{\$82{,}500} = 5.8182 \end{aligned}$$

From the interest factor tables for $n = 8$: $(P/A,\, 7\%,\, 8) = 5.9713$ and $(P/A,\, 8\%,\, 8) = 5.7466$. This yields an initial estimate of $i_0 \approx 7.68\%$, suggesting 8% as our first trial

rate.[2]

4. **Refinement and Bracketing near i_0:**

   - *Trial 1 at $i = 8\%$:*

$$\begin{aligned} PW(8\%) &= -400{,}000 + 75{,}000\,(P/A,\,8\%,\,8) - 80{,}000\,(P/F,\,8\%,\,5) + 60{,}000\,(P/F,\,8\%,\,8) \\ &= -400{,}000 + 75{,}000(5.7466) - 80{,}000(0.6806) + 60{,}000(0.5403) \\ &= -400{,}000 + 430{,}995.00 - 54{,}448.00 + 32{,}418.00 \\ &= \$8{,}965.00 > 0 \end{aligned}$$

   Since $PW(8\%) > 0$, the return is higher than 8%.

   - *Trial 2 at $i = 10\%$:*

$$\begin{aligned} PW(10\%) &= -400{,}000 + 75{,}000\,(P/A,\,10\%,\,8) - 80{,}000\,(P/F,\,10\%,\,5) + 60{,}000\,(P/F,\,10\%,\,8) \\ &= -400{,}000 + 75{,}000(5.3349) - 80{,}000(0.6209) + 60{,}000(0.4665) \\ &= -400{,}000 + 400{,}117.50 - 49{,}672.00 + 27{,}990.00 \\ &= -\$21{,}564.50 < 0 \end{aligned}$$

5. **Linear Interpolation between 8% and 10%:**

$$\begin{aligned} i^* &\approx 8\% + \frac{\$8{,}965.00}{\$8{,}965.00 - (-\$21{,}564.50)}\,(2\%) \\ &= 8\% + \frac{8{,}965.00}{30{,}529.50}\,(2\%) \\ &= 8\% + 0.59\% \\ &= 8.59\% \end{aligned}$$

   (Exact Excel value: `=IRR` $= 8.56\%$.)

6. **Decision under MARR = 12%:** Evaluating at the MARR:

$$\begin{aligned} PW(12\%) &= -400{,}000 + 75{,}000(4.9676) - 80{,}000(0.5674) + 60{,}000(0.4039) \\ &= -400{,}000 + 372{,}570.00 - 45{,}392.00 + 24{,}234.00 \\ &= -\$48{,}588.00 < 0 \end{aligned}$$

   Because $i^* \approx 8.59\% <$ MARR $= 12\%$ (and $PW(12\%) = -\$48{,}588.00 < 0$), the project remains **not acceptable**.

[2] If a student alternatively converted all receipts into a single future sum at Year 8 ($F = 8(\$75{,}000) + \$60{,}000 = \$660{,}000$), $(P/F, i_0, 8) = 480{,}000/660{,}000 = 0.7273 \Rightarrow i_0 \approx 4.06\%$. Converting to uniform A is vastly superior because it preserves the dominant annuity structure of the cash flows.

Part 11. *Descartes' Rule and Norstrom's Criterion Evaluation:*

- **Raw Cash Flows:** $[-\$400k, +\$75k, +\$75k, +\$75k, +\$75k, -\$5k, +\$75k, +\$75k, +\$135k]$. Sign changes occur from Year 0 to 1 ($-$ to $+$), Year 4 to 5 ($+$ to $-$), and Year 5 to 6 ($-$ to $+$). With 3 sign changes, Descartes' rule states there are at most 3 real roots with $i > -100\%$.
- **Cumulative Series (S_t):**

t	0	1	2	3	4	5	6	7	8
S_t	$-\$400k$	$-\$325k$	$-\$250k$	$-\$175k$	$-\$100k$	$-\$105k$	$-\$30k$	$+\$45k$	$+\$180k$

Because $S_0 = -\$400k < 0$ and the cumulative series changes sign exactly once (between Year 6 and Year 7), **Norstrom's criterion is fully satisfied**. This mathematically guarantees that $i^* = 8.56\%$ is the unique positive internal rate of return.

Exercise 2: Rooftop Solar Array for a Regional Hospital

Solution:

Given: Initial cost \$480,000; base savings $A_1 = \$62{,}000$ rising by $G = \$9{,}000$/yr over $n = 10$ years; salvage value $S = \$40{,}000$; MARR = 12%. **Find:** i^* and the investment decision.

Part 1. Complete Present Worth equation in standard factor notation:

$$PW(i^*) = -480{,}000 + 62{,}000\,(P/A,\, i^*,\, 10) + 9{,}000\,(P/G,\, i^*,\, 10) + 40{,}000\,(P/F,\, i^*,\, 10) = 0$$

Part 2. *Initial guess and choice of transformation:*

Approach 1: Collapsing into a Single Terminal Future Sum (F at $t = 10$):

$$\begin{aligned} F &= 10(\$62{,}000) + 9{,}000\left(\sum_{t=0}^{9} t\right) + 40{,}000 \\ &= 620{,}000 + 9{,}000(45) + 40{,}000 \\ &= \$1{,}065{,}000 \\ (P/F,\, i_0,\, 10) &= \frac{\$480{,}000}{\$1{,}065{,}000} \\ &= 0.4507 \\ i_0 &= \left(\frac{\$1{,}065{,}000}{\$480{,}000}\right)^{1/10} - 1 \approx 8.30\% \end{aligned}$$

Directional Bias: Shifting early savings (Years 1–9) to Year 10 without compounding understates their true present earning power. Therefore, $i_0 = 8.30\%$ is guaranteed to be a conservative lower bound. Testing $PW(8.30\%)$:

$$\begin{aligned} PW(8.30\%) &= -480{,}000 + 62{,}000(6.6202) + 9{,}000(25.4820) + 40{,}000(0.4505) \\ &= \$177{,}810.40 > 0 \end{aligned}$$

Since $PW > 0$, the true root is significantly above 8.30%.

Approach 2 (Alternative): Converting into an Equivalent Uniform Annual Series (A'): Instead of collapsing all cash flows to Year 10, an alternative transformation converts the receipts into an equivalent undiscounted uniform annual series A':

- **Base savings:** $A_1 = \$62{,}000/\text{year}$.
- **Average annual gradient:** With an undiscounted average factor $(A/G, 0\%, 10) = \frac{n-1}{2} = 4.5$,

$$A_G = \$9{,}000 \times 4.5 = \$40{,}500/\text{year}$$

- **Annualized salvage:**

$$A_S = \frac{\$40{,}000}{10} = \$4{,}000/\text{year}$$

- **Total equivalent annuity:**

$$A' = \$62{,}000 + \$40{,}500 + \$4{,}000 = \$106{,}500/\text{year}$$

Equating $PW_D = PW_R$ to find the (P/A) factor:

$$\begin{aligned} \$480{,}000 &= \$106{,}500\,(P/A,\, i_0,\, 10) \\ (P/A,\, i_0,\, 10) &= \frac{\$480{,}000}{\$106{,}500} = 4.5070 \end{aligned}$$

From the interest factor tables for $n = 10$, $(P/A,\, 16\%,\, 10) = 4.8332$ and $(P/A,\, 18\%,\, 10) = 4.4941$, which interpolates to $i_0 \approx 17.92\%$.

Pedagogical Comparison of What to Transform:

- **Future Sum (F):** Shifting all cash flows to Year 10 delays early savings, understating their earning power and yielding a **lower bound** ($i_0 = 8.30\%$, error of -6.42%).

- **Uniform Series (A'):** Averaging the increasing gradient evenly across all years shifts future revenue forward in time, overstating early earning power and yielding an **upper bound** ($i_0 = 17.92\%$, error of $+3.20\%$).
- **The Smart Bracket:** Without any discounting, the two transformations bracket the root immediately: $8.30\% \le i^* \le 17.92\%$. The midpoint is $\frac{8.30\%+17.92\%}{2} = 13.11\%$, landing right next to the true root (14.72%).

Part 3. Moving up to trial rates 2% apart (14% and 16%):

$$\begin{aligned} PW(14\%) &= -480{,}000 + 62{,}000(5.2161) + 9{,}000(17.9906) + 40{,}000(0.2697) \\ &= \$16{,}101.60 > 0 \\ PW(16\%) &= -480{,}000 + 62{,}000(4.8332) + 9{,}000(16.0399) + 40{,}000(0.2267) \\ &= -\$26{,}914.50 < 0 \end{aligned}$$

Linear interpolation between 14% and 16%:

$$\begin{aligned} i^* &\approx 14\% + \frac{\$16{,}101.60}{\$16{,}101.60 - (-\$26{,}914.50)}\,(2\%) \\ &= 14\% + \frac{16{,}101.60}{43{,}016.10}\,(2\%) \\ &= 14.75\% \end{aligned}$$

(Exact Excel value: `=IRR` $= 14.72\%$.)

Part 4. Since $i^* \approx 14.75\% \ge$ MARR $= 12\%$, the hospital **should invest** in the solar array. Verification at the MARR:

$$\begin{aligned} PW(12\%) &= -480{,}000 + 62{,}000(5.6502) + 9{,}000(20.2541) + 40{,}000(0.3220) \\ &= \$65{,}479.30 > 0 \end{aligned}$$

Exercise 3: Geothermal Power Station Acquisition

Solution:

Given: First cost \$850,000; revenue $A_1 = \$210{,}000$ declining geometrically at $g = -7\%$ per year for $n = 9$ years; no salvage value; MARR $= 16\%$. **Find:** i^* and the decision.

Part 1. Present Worth equivalence equation:

$$PW(i^*) = -850{,}000 + 210{,}000\,(P/A,\ -7\%,\ i^*,\ 9) = 0$$

From Course Notes Chapter 2, the analytical formula for the geometric gradient factor is:

$$(P/A,\, g,\, i,\, n) = \frac{\left(\frac{1+g}{1+i}\right)^n - 1}{g - i} \quad (g \neq i)$$

Sign Rule Note: With $g = -0.07$, the denominator is $(-0.07 - i^*)$. Both numerator and denominator are negative, yielding a positive factor.

Part 2. *Initial estimate and comparison of transformations:*

Approach 1: Collapsing into a Single Terminal Future Sum (F at $t = 9$): Neglecting the time value of money, the 9 revenues sum to:

$$F = \sum_{t=0}^{8} 210{,}000(1 - 0.07)^t = 210{,}000 \left(\frac{1 - 0.93^9}{0.07}\right) = \$1{,}438{,}767$$

$$(P/F,\, i_0,\, 9) = \frac{\$850{,}000}{\$1{,}438{,}767} = 0.5908$$

$$i_0 = \left(\frac{\$1{,}438{,}767}{\$850{,}000}\right)^{1/9} - 1 \approx 6.02\%$$

Evaluating $PW(6.02\%)$ using $(P/A,\, -7\%,\, 6.02\%,\, 9) = 5.3187$:

$$PW(6.02\%) = -850{,}000 + 210{,}000(5.3187) = \$266{,}927.00 > 0$$

Because revenues decline geometrically ($g = -7\%$), receipts are heavily front-loaded (\$210,000 in Year 1 down to \$111,387 in Year 9). Pushing all revenue to Year 9 delays early receipts by up to 8 years, severely understating earning power and resulting in an estimate (6.02%) that is far below the true rate (13.64%).

Approach 2 (Alternative): Converting into an Equivalent Uniform Series (A_{avg}): A much smarter transformation converts the total undiscounted revenue into an average annual receipt:

$$A_{\text{avg}} = \frac{\$1{,}438{,}767}{9}$$
$$= \$159{,}863/\text{year}$$

Equating $PW_D = PW_R$ to find the (P/A) factor:

$$\begin{aligned} \$850{,}000 &= \$159{,}863\,(P/A,\, i_0,\, 9) \\ (P/A,\, i_0,\, 9) &= \frac{\$850{,}000}{\$159{,}863} \\ &= 5.3171 \end{aligned}$$

From the interest factor tables for $n = 9$:

$$\begin{aligned} (P/A,\, 12\%,\, 9) &= 5.3282 \\ (P/A,\, 14\%,\, 9) &= 4.9464 \end{aligned}$$

Because $(P/A,\, 12\%,\, 9) = 5.3282 \approx 5.3171$, this yields $i_0 \approx 12.06\%$ (an error of only 1.58%, slashing estimation error by nearly 80%!).

Key Pedagogical Takeaway: Converting into a uniform series (A) preserves the multi-year distribution of receipts much more accurately than a terminal lump sum (F). Furthermore, $i_0 \approx 12.06\%$ points the student immediately to **12%** as their first trial, bracketing the root (13.64%) directly between 12% and 14% in a single step!

Part 3. Bracketing between 12% and 14%:

$$\begin{aligned} PW(12\%) &= -850{,}000 + 210{,}000(4.2754) \\ &= \$47{,}834.00 > 0 \\ PW(14\%) &= -850{,}000 + 210{,}000(3.9999) \\ &= -\$10{,}021.00 < 0 \end{aligned}$$

Linear interpolation:

$$\begin{aligned} i^* &\approx 12\% + \frac{\$47{,}834.00}{\$47{,}834.00 - (-\$10{,}021.00)}\,(2\%) \\ &= 12\% + \frac{47{,}834.00}{57{,}855.00}\,(2\%) \\ &= 13.65\% \end{aligned}$$

(Exact Excel value: `=IRR` $= 13.64\%$.)

Part 4. Since $i^* \approx 13.65\% <$ MARR $= 16\%$, the buyer **should not accept** the asking

price of \$850,000. Check at MARR:

$$\begin{aligned} PW(16\%) &= -850{,}000 + 210{,}000(3.7529) \\ &= -\$61{,}891.00 < 0 \end{aligned}$$

Exercise 4: Equipment Purchase for a University Makerspace (Excel Analysis)

Solution:

Given: First cost \$250,000; salvage \$60,000 at Year 6; MARR = 12%.

Part 1. *Scenario 1:* Uniform annual savings of \$55,000 per year. Because the cash flow consists of a single outlay P, a uniform annuity A, and a single terminal salvage F, Excel's `RATE` function applies directly:

$$\texttt{=RATE(6, 55000, -250000, 60000)} \;\Rightarrow\; i^* = 12.83\%$$

Since $i^* = 12.83\% \geq \text{MARR} = 12\%$, the makerspace should **accept** the purchase. Check at MARR: $PW(12\%) = -250{,}000 + 55{,}000(4.1114) + 60{,}000(0.5066) = \$6{,}525.27 > 0$.

Part 2. *Scenario 2:* Arithmetic gradient savings: The annual savings are \$40,000 + $(t-1)$(\$6,000). At Year 6, the cash flow combines the operational savings of \$40,000 + 5(\$6,000) = \$70,000 and the \$60,000 salvage value, yielding \$130,000:

Cell	Year (t)	Net Cash Flow
B2	0	−\$250,000
B3	1	\$40,000
B4	2	\$46,000
B5	3	\$52,000
B6	4	\$58,000
B7	5	\$64,000
B8	6	\$130,000

The Excel function syntax is:

$$\texttt{=IRR(B2:B8)} \;\Rightarrow\; i^* = 11.82\%$$

Since $i^* = 11.82\% < \text{MARR} = 12\%$, the makerspace should **reject** the purchase. Check at MARR: $PW(12\%) = -250{,}000 + 40{,}000(4.1114) + 6{,}000(8.9447) + 60{,}000(0.5066) = -\$1{,}564.81 < 0$.

Part 3. `RATE` strictly requires a single constant payment amount (`pmt`) for all intermediate periods. The arithmetic gradient produces varying annual savings, so they cannot be represented by a single `pmt`. Instead, the cash flows must be entered period-by-period into contiguous cells and solved via `IRR`.

Part B — Multiple Alternatives (Exercises 5–7)

Exercise 5: Airport Shuttle Fleet for a Hotel Group

Solution:

Given: Diesel (A): First cost \$240,000; income \$52,000 in Year 1 rising by \$4,000/yr; salvage \$30,000. Electric (B): First cost \$410,000; income \$91,000 in Year 1 rising by \$4,000/yr; battery replacement \$45,000 at $t = 4$; salvage \$60,000. Both last 8 years; MARR = 10%.

Part 1. Year-by-year cash flows:

Year (t)	Diesel Fleet (A)	Electric Fleet (B)
0	−\$240,000	−\$410,000
1	\$52,000	\$91,000
2	\$56,000	\$95,000
3	\$60,000	\$99,000
4	\$64,000	\$58,000
5	\$68,000	\$107,000
6	\$72,000	\$111,000
7	\$76,000	\$115,000
8	\$110,000	\$179,000

Screening individual RORs using Course Notes heuristics in steps:

- **Diesel Fleet (A):**

  a. *Undiscounted equivalent annuity:*

$$A'_A = \$52{,}000 + \$4{,}000\left(\frac{8-1}{2}\right) + \frac{\$30{,}000}{8}$$
$$= \$52{,}000 + \$14{,}000 + \$3{,}750 = \$69{,}750/\text{year}$$

  b. *Initial factor and estimate:*

$$\$240{,}000 = \$69{,}750\,(P/A,\, i_0,\, 8)$$
$$(P/A,\, i_0,\, 8) = \frac{\$240{,}000}{\$69{,}750} = 3.4409$$

From factor tables for $n = 8$, $(P/A,\ 20\%,\ 8) = 3.8372$ and $(P/A,\ 25\%,\ 8) = 3.3289$. This indicates $i_0 \approx 23.8\%$, directing us to test 20% and 22%.

c. *Trial evaluations and interpolation:*

$$\begin{aligned} PW_A(20\%) &= -240{,}000 + 52{,}000(3.8372) + 4{,}000(9.8831) + 30{,}000(0.2326) \\ &= \$6{,}044.80 > 0 \end{aligned}$$

$$\begin{aligned} PW_A(22\%) &= -240{,}000 + 52{,}000(3.6193) + 4{,}000(9.0417) + 30{,}000(0.2038) \\ &= -\$9{,}515.60 < 0 \end{aligned}$$

$$i_A^* \approx 20\% + \frac{\$6{,}044.80}{\$6{,}044.80 - (-\$9{,}515.60)}\,(2\%) = 20.78\% \quad \text{(Excel: 20.75\%)}$$

- **Electric Fleet (B):**

a. *Undiscounted disbursements and equivalent annuity:*

$$\begin{aligned} PW_D &= \$410{,}000 + \$45{,}000 = \$455{,}000 \\ A'_B &= \$91{,}000 + \$4{,}000\left(\frac{8-1}{2}\right) + \frac{\$60{,}000}{8} \\ &= \$91{,}000 + \$14{,}000 + \$7{,}500 = \$112{,}500/\text{year} \end{aligned}$$

b. *Initial factor and estimate:*

$$\begin{aligned} \$455{,}000 &= \$112{,}500\,(P/A,\ i_0,\ 8) \\ (P/A,\ i_0,\ 8) &= \frac{\$455{,}000}{\$112{,}500} = 4.0444 \end{aligned}$$

From factor tables for $n = 8$, $(P/A,\ 16\%,\ 8) = 4.3436$ and $(P/A,\ 18\%,\ 8) = 4.0776 \approx 4.0444$, indicating $i_0 \approx 18.26\%$. This points directly to trial rates of 16% and 18%.

c. *Trial evaluations and interpolation:*

$$\begin{aligned} PW_B(16\%) &= -410{,}000 + 91{,}000(4.3436) + 4{,}000(11.8962) - 45{,}000(0.5523) + 60{,}000(0.3050) \\ &= \$26{,}298.90 > 0 \end{aligned}$$

$$\begin{aligned} PW_B(18\%) &= -410{,}000 + 91{,}000(4.0776) + 4{,}000(10.8292) - 45{,}000(0.5158) + 60{,}000(0.2660) \\ &= -\$2{,}872.60 < 0 \end{aligned}$$

$$i_B^* \approx 16\% + \frac{\$26{,}298.90}{\$26{,}298.90 - (-\$2{,}872.60)}\,(2\%) = 17.80\% \quad \text{(Excel: 17.79\%)}$$

Both $i_A^* \approx 20.78\%$ and $i_B^* \approx 17.80\%$ exceed MARR = 10%. Both options are viable on

their own.

Part 2. Incremental cash flows $(B - A)$: Notice that both options share an identical annual gradient of \$4,000/year. When taking the difference:

$$\Delta CF_t = [91{,}000 + 4{,}000(t-1)] - [52{,}000 + 4{,}000(t-1)] = \$39{,}000$$

The gradient cancels out completely! The incremental series consists of an extra outlay of \$170,000, a uniform annual income difference of \$39,000, the Year 4 battery replacement of −\$45,000, and the salvage difference of +\$30,000:

Year (t)	**Diesel (A)**	**Electric (B)**	**Incremental ($B - A$)**
0	−\$240,000	−\$410,000	−\$170,000
1–3	$\$52{,}000 + 4{,}000(t-1)$	$\$91{,}000 + 4{,}000(t-1)$	\$39,000 each year
4	\$64,000	\$58,000	−\$6,000
5–7	$\$68{,}000 + 4{,}000(t-5)$	$\$107{,}000 + 4{,}000(t-5)$	\$39,000 each year
8	\$110,000	\$179,000	\$69,000

Part 3. Incremental rate of return Δi^*_{B-A}:

a. *Undiscounted incremental conversion:*

$$\Delta PW_D = \$170{,}000 + \$45{,}000 = \$215{,}000$$

$$\Delta A' = \$39{,}000 + \frac{\$30{,}000}{8} = \$42{,}750/\text{year}$$

$$(P/A,\, i_0,\, 8) = \frac{\$215{,}000}{\$42{,}750} = 5.0292$$

From factor tables for $n = 8$, $(P/A,\, 10\%,\, 8) = 5.3349$ and $(P/A,\, 12\%,\, 8) = 4.9676$. This yields $i_0 \approx 11.65\%$, directing us to bracket with 12% and 14%.

b. *Trial evaluations and interpolation:*

$$PW_\Delta(i) = -170{,}000 + 39{,}000\,(P/A,\, i,\, 8) - 45{,}000\,(P/F,\, i,\, 4) + 30{,}000\,(P/F,\, i,\, 8) = 0$$

$$\begin{aligned} PW_\Delta(12\%) &= -170{,}000 + 39{,}000(4.9676) - 45{,}000(0.6355) + 30{,}000(0.4039) \\ &= \$7{,}255.90 > 0 \end{aligned}$$

$$\begin{aligned} PW_\Delta(14\%) &= -170{,}000 + 39{,}000(4.6389) - 45{,}000(0.5921) + 30{,}000(0.3506) \\ &= -\$5{,}209.40 < 0 \end{aligned}$$

Linear interpolation:

$$\Delta i^*_{B-A} \approx 12\% + \frac{\$7{,}255.90}{\$7{,}255.90 - (-\$5{,}209.40)}\,(2\%) = 13.16\% \quad \text{(Excel: } 13.13\%\text{)}$$

Part 4. Since $\Delta i^*_{B-A} = 13.13\% \geq \text{MARR} = 10\%$, the additional investment of \$170,000 in the electric fleet earns more than the hurdle rate: **choose the electric fleet (B)**. *Pedagogical Takeaway:* Although Diesel (A) has a higher individual ROR (20.75% vs. 17.79%), its return is earned on a smaller capital base. As established in Course Notes Chapter 8, selecting the option with the highest individual ROR is a classic error; what governs the decision is whether the *incremental* capital earns at least the MARR.

Part 5. Sensitivity Analysis on Battery Replacement Cost (R): At MARR = 10%, the base incremental Present Worth is $PW_\Delta(10\%) = \$21{,}321.74$. If the battery replacement at Year 4 costs R, the revised incremental Present Worth is:

$$\begin{aligned} PW_\Delta(R) &= \$21{,}321.74 - (R - 45{,}000)\,(P/F,\ 10\%,\ 4) \geq 0 \\ R &\leq 45{,}000 + \frac{\$21{,}321.74}{(1.10)^{-4}} \\ &= 45{,}000 + \frac{\$21{,}321.74}{0.683013} \\ &= \$76{,}217.16 \end{aligned}$$

The hotel group is indifferent at a replacement cost of **\$76,217.16**. If the battery exceeds this price, Diesel (A) becomes the superior economic choice.

Exercise 6: Gaming Rigs for a University Esports Arena

Solution:

Given: Option A (life 2 yr): First cost \$48,000; revenue \$24,000 in Year 1, \$26,000 in Year 2; salvage \$9,000. Option B (life 3 yr): First cost \$85,000; revenue \$30,000 in Year 1 rising by \$2,000/yr; salvage \$14,000. MARR = 12%.

Part 1. Cash flows over each option's own life cycle:

Year	Option A (2 yr)	Option B (3 yr)
0	−\$48,000	−\$85,000
1	\$24,000	\$30,000
2	\$26,000 + \$9,000 = \$35,000	\$32,000
3	—	\$34,000 + \$14,000 = \$48,000

Part 2. Individual screening ROR using Course Notes heuristics in steps:

- **Option A (2-year life):**
  a. *Undiscounted equivalent annuity:* Total undiscounted receipts over 2 years: \$24,000+

$\$26{,}000 + \$9{,}000 = \$59{,}000$.

$$A'_A = \frac{\$59{,}000}{2} = \$29{,}500/\text{year}$$

$$(P/A,\, i_0,\, 2) = \frac{\$48{,}000}{\$29{,}500} = 1.6271$$

From factor tables for $n = 2$, $(P/A,\, 12\%,\, 2) = 1.6901$ and $(P/A,\, 14\%,\, 2) = 1.6467$. This indicates $i_0 \approx 14.9\%$, directing us to test 12% and 14%.

b. *Trial evaluations and interpolation:*

$$\begin{aligned} PW_A(12\%) &= -48{,}000 + 24{,}000(1.6901) + 2{,}000(0.7972) + 9{,}000(0.7972) \\ &= \$1{,}331.60 > 0 \end{aligned}$$

$$\begin{aligned} PW_A(14\%) &= -48{,}000 + 24{,}000(1.6467) + 2{,}000(0.7695) + 9{,}000(0.7695) \\ &= -\$14.70 < 0 \end{aligned}$$

$$i^*_A \approx 12\% + \frac{\$1{,}331.60}{\$1{,}331.60 - (-\$14.70)}\,(2\%) = 13.98\% \quad \text{(Excel: 13.98\%)}$$

- **Option B (3-year life):**

a. *Undiscounted equivalent annuity:* Total undiscounted receipts over 3 years: $\$30{,}000 + \$32{,}000 + \$34{,}000 + \$14{,}000 = \$110{,}000$.

$$A'_B = \frac{\$110{,}000}{3} = \$36{,}667/\text{year}$$

$$(P/A,\, i_0,\, 3) = \frac{\$85{,}000}{\$36{,}667} = 2.3182$$

From factor tables for $n = 3$, $(P/A,\, 12\%,\, 3) = 2.4018$ and $(P/A,\, 14\%,\, 3) = 2.3216 \approx 2.3182$. This indicates $i_0 \approx 14.1\%$, directing us to test 12% and 14%.

b. *Trial evaluations and interpolation:*

$$\begin{aligned} PW_B(12\%) &= -85{,}000 + 30{,}000(2.4018) + 2{,}000(2.2208) + 14{,}000(0.7118) \\ &= \$1{,}460.80 > 0 \end{aligned}$$

$$\begin{aligned} PW_B(14\%) &= -85{,}000 + 30{,}000(2.3216) + 2{,}000(2.1194) + 14{,}000(0.6750) \\ &= -\$1{,}663.20 < 0 \end{aligned}$$

$$i^*_B \approx 12\% + \frac{\$1{,}460.80}{\$1{,}460.80 - (-\$1{,}663.20)}\,(2\%) = 12.94\% \quad \text{(Excel: 12.92\%)}$$

Both exceed 12%. Both options pass initial screening.

Part 3. Because the lives differ (2 years vs. 3 years), the equal-service study period for Present Worth analysis is the Least Common Multiple: $\text{LCM}(2,3) = \mathbf{6}$ years. Option A is renewed at Years 0, 2, and 4 (3 cycles); Option B is renewed at Years 0 and 3 (2 cycles).

Part 4. Extended cash flows over the 6-year study period:

Year	Option A (3 cycles)	Option B (2 cycles)	Incremental ($B-A$)
0	−\$48,000	−\$85,000	−\$37,000
1	\$24,000	\$30,000	\$6,000
2	\$26,000 + \$9,000 − \$48,000 = −\$13,000	\$32,000	\$45,000
3	\$24,000	\$34,000 + \$14,000 − \$85,000 = −\$37,000	−\$61,000
4	\$26,000 + \$9,000 − \$48,000 = −\$13,000	\$30,000	\$43,000
5	\$24,000	\$32,000	\$8,000
6	\$26,000 + \$9,000 = \$35,000	\$34,000 + \$14,000 = \$48,000	\$13,000

Part 5. Incremental Present Worth equation over 6 years:

$$\begin{aligned} PW_\Delta(10\%) &= -37{,}000 + 6{,}000(0.9091) + 45{,}000(0.8264) - 61{,}000(0.7513) \\ &\quad + 43{,}000(0.6830) + 8{,}000(0.6209) + 13{,}000(0.5645) \\ &= \$1{,}488.00 > 0 \end{aligned}$$

$$\begin{aligned} PW_\Delta(12\%) &= -37{,}000 + 6{,}000(0.8929) + 45{,}000(0.7972) - 61{,}000(0.7118) \\ &\quad + 43{,}000(0.6355) + 8{,}000(0.5674) + 13{,}000(0.5066) \\ &= -\$736.90 < 0 \end{aligned}$$

Linear interpolation:

$$\Delta i^*_{B-A} \approx 10\% + \frac{\$1{,}488.00}{\$1{,}488.00 - (-\$736.90)}\,(2\%) = 11.34\% \quad \text{(Excel: } 11.32\%\text{)}$$

Part 6. Since $\Delta i^*_{B-A} = 11.32\% < \text{MARR} = 12\%$, the extra investment of \$37,000 for the high-end rigs is not economically justified: **select Option A (mid-range PCs)**.

Part 7. *Annual Worth (AW) Verification over a Single Life Cycle:*

$$\begin{aligned} AW_A(12\%) &= -48{,}000\,(A/P,\,12\%,\,2) + 24{,}000 + 2{,}000\,(A/G,\,12\%,\,2) + 9{,}000\,(A/F,\,12\%,\,2) \\ &= -48{,}000(0.5917) + 24{,}000 + 2{,}000(0.4717) + 9{,}000(0.4717) \\ &= \$787.10 \end{aligned}$$

$$\begin{aligned} AW_B(12\%) &= -85{,}000\,(A/P,\,12\%,\,3) + 30{,}000 + 2{,}000\,(A/G,\,12\%,\,3) + 14{,}000\,(A/F,\,12\%,\,3) \\ &= -85{,}000(0.4163) + 30{,}000 + 2{,}000(0.9246) + 14{,}000(0.2963) \\ &= \$611.90 \end{aligned}$$

(Note: Full formula precision yields $AW_B = \$608.44$; table factor rounding yields \$611.90.) Since $AW_A(12\%) > AW_B(12\%)$, Option A is unequivocally confirmed as superior. *Pedagogical Insight:* Because the Annual Worth of one cycle equals the Annual Worth over any integer number of cycles, the equal-service requirement is satisfied automatically. Evaluating AW directly avoids expanding cash flows over the LCM and bypasses the multiple sign changes that typically arise in multi-cycle incremental cash flows.

Exercise 7: Automation Packages for a Distribution Center

Solution:

Given: Packages A (\$120k), B (\$210k), C (\$260k), D (\$380k). All last $n = 6$ years; MARR = 12%.

Part 1. *Individual Screening Rates of Return (Course Notes Section 7.2.1 Heuristic):*

Applying the manual estimation procedure, each package's terminal salvage S is converted to an undiscounted equivalent annuity component $S/n = S/6$ added to annual savings A to form $A' = A + S/6$. The initial trial rate i_0 is found from $(P/A,\, i_0,\, 6) = P/A'$, followed by bracketing and linear interpolation:

- **Package A** ($P = \$120{,}000$, $A = \$40{,}000$, $S = \$10{,}000$):

$$A'_A = 40{,}000 + \frac{10{,}000}{6} = \$41{,}667/\text{year}$$

$$(P/A,\, i_0,\, 6) = \frac{\$120{,}000}{\$41{,}667} = 2.8800 \implies i_0 \approx 26.1\%$$

Evaluating $PW_A(i)$ at 24% and 26%:

$$\begin{aligned} PW_A(24\%) &= -120{,}000 + 40{,}000\,(P/A,\, 24\%,\, 6) + 10{,}000\,(P/F,\, 24\%,\, 6) \\ &= -120{,}000 + 40{,}000(3.0205) + 10{,}000(0.2751) \\ &= \$3{,}571.00 > 0 \end{aligned}$$

$$\begin{aligned} PW_A(26\%) &= -120{,}000 + 40{,}000\,(P/A,\, 26\%,\, 6) + 10{,}000\,(P/F,\, 26\%,\, 6) \\ &= -120{,}000 + 40{,}000(2.8850) + 10{,}000(0.2499) \\ &= -\$2{,}101.00 < 0 \end{aligned}$$

$$\begin{aligned} i^*_A &\approx 24\% + \frac{\$3{,}571.00}{\$3{,}571.00 - (-\$2{,}101.00)}\,(2\%) \\ &= 24\% + \frac{\$3{,}571.00}{\$5{,}672.00}\,(2\%) \\ &= 25.26\% \quad \text{(Excel: } 25.24\%\text{)} \end{aligned}$$

- **Package B** ($P = \$210{,}000$, $A = \$45{,}000$, $S = \$15{,}000$):

$$A'_B = 45{,}000 + \frac{15{,}000}{6} = \$47{,}500/\text{year}$$

$$(P/A,\, i_0,\, 6) = \frac{\$210{,}000}{\$47{,}500} = 4.4211 \implies i_0 \approx 9.5\%$$

Evaluating $PW_B(i)$ at 8% and 10%:

$$\begin{aligned} PW_B(8\%) &= -210{,}000 + 45{,}000\,(P/A,\, 8\%,\, 6) + 15{,}000\,(P/F,\, 8\%,\, 6) \\ &= -210{,}000 + 45{,}000(4.6229) + 15{,}000(0.6302) \\ &= \$7{,}483.50 > 0 \end{aligned}$$

$$\begin{aligned} PW_B(10\%) &= -210{,}000 + 45{,}000\,(P/A,\, 10\%,\, 6) + 15{,}000\,(P/F,\, 10\%,\, 6) \\ &= -210{,}000 + 45{,}000(4.3553) + 15{,}000(0.5645) \\ &= -\$5{,}544.00 < 0 \end{aligned}$$

$$\begin{aligned} i_B^* &\approx 8\% + \frac{\$7{,}483.50}{\$7{,}483.50 - (-\$5{,}544.00)}\,(2\%) \\ &= 8\% + \frac{\$7{,}483.50}{\$13{,}027.50}\,(2\%) \\ &= 9.15\% \quad \text{(Excel: } 9.12\%\text{)} \end{aligned}$$

- **Package C** ($P = \$260{,}000$, $A = \$75{,}000$, $S = \$25{,}000$):

$$A'_C = 75{,}000 + \frac{25{,}000}{6} = \$79{,}167/\text{year}$$

$$(P/A,\, i_0,\, 6) = \frac{\$260{,}000}{\$79{,}167} = 3.2842 \implies i_0 \approx 20.5\%$$

Evaluating $PW_C(i)$ at 18% and 20%:

$$\begin{aligned} PW_C(18\%) &= -260{,}000 + 75{,}000\,(P/A,\,18\%,\,6) + 25{,}000\,(P/F,\,18\%,\,6) \\ &= -260{,}000 + 75{,}000(3.4976) + 25{,}000(0.3704) \\ &= \$11{,}580.00 > 0 \end{aligned}$$

$$\begin{aligned} PW_C(20\%) &= -260{,}000 + 75{,}000\,(P/A,\,20\%,\,6) + 25{,}000\,(P/F,\,20\%,\,6) \\ &= -260{,}000 + 75{,}000(3.3255) + 25{,}000(0.3349) \\ &= -\$2{,}215.00 < 0 \end{aligned}$$

$$\begin{aligned} i_C^* &\approx 18\% + \frac{\$11{,}580.00}{\$11{,}580.00 - (-\$2{,}215.00)}\,(2\%) \\ &= 18\% + \frac{\$11{,}580.00}{\$13{,}795.00}\,(2\%) \\ &= 19.68\% \quad \text{(Excel: } 19.67\%\text{)} \end{aligned}$$

- **Package D** ($P = \$380{,}000$, $A = \$100{,}000$, $S = \$40{,}000$):

$$A'_D = 100{,}000 + \frac{40{,}000}{6} = \$106{,}667/\text{year}$$

$$(P/A,\,i_0,\,6) = \frac{\$380{,}000}{\$106{,}667} = 3.5625 \implies i_0 \approx 17.3\%$$

Evaluating $PW_D(i)$ at 16% and 18%:

$$\begin{aligned} PW_D(16\%) &= -380{,}000 + 100{,}000\,(P/A,\,16\%,\,6) + 40{,}000\,(P/F,\,16\%,\,6) \\ &= -380{,}000 + 100{,}000(3.6847) + 40{,}000(0.4104) \\ &= \$4{,}886.00 > 0 \end{aligned}$$

$$\begin{aligned} PW_D(18\%) &= -380{,}000 + 100{,}000\,(P/A,\,18\%,\,6) + 40{,}000\,(P/F,\,18\%,\,6) \\ &= -380{,}000 + 100{,}000(3.4976) + 40{,}000(0.3704) \\ &= -\$15{,}424.00 < 0 \end{aligned}$$

$$\begin{aligned} i_D^* &\approx 16\% + \frac{\$4{,}886.00}{\$4{,}886.00 - (-\$15{,}424.00)}\,(2\%) \\ &= 16\% + \frac{\$4{,}886.00}{\$20{,}310.00}\,(2\%) \\ &= 16.48\% \quad \text{(Excel: } 16.47\%\text{)} \end{aligned}$$

Package	First Cost	Net Annual Savings	Interpolated i^*	Status at MARR = 12%
A	\$120,000	\$40,000	25.26% (Excel: 25.24%)	Viable (Defender)
B	\$210,000	\$45,000	9.15% (Excel: 9.12%)	**Eliminated** ($i^* <$ MARR)
C	\$260,000	\$75,000	19.68% (Excel: 19.67%)	Viable (Challenger 1)
D	\$380,000	\$100,000	16.48% (Excel: 16.47%)	Viable (Challenger 2)

Package A has the highest individual i^* (25.26%). Package B is eliminated because $i^* = 9.15\% < 12\%$. The surviving alternatives in order of increasing initial investment are A, C, and D.

Part 2. *Systematic Pairwise Incremental Analysis:*

Comparison 1: Challenger C vs. Defender A

$$\Delta P = 260{,}000 - 120{,}000 = \$140{,}000$$
$$\Delta A = 75{,}000 - 40{,}000 = \$35{,}000$$
$$\Delta S = 25{,}000 - 10{,}000 = \$15{,}000$$
$$\Delta A' = 35{,}000 + \frac{15{,}000}{6} = \$37{,}500/\text{year}$$
$$(P/A,\, i_0,\, 6) = \frac{\$140{,}000}{\$37{,}500} = 3.7333 \implies i_0 \approx 15.5\%$$

Evaluating incremental Present Worth $PW_\Delta(i)$ at 14% and 16%:

$$\begin{aligned} PW_\Delta(14\%) &= -140{,}000 + 35{,}000\,(P/A,\, 14\%,\, 6) + 15{,}000\,(P/F,\, 14\%,\, 6) \\ &= -140{,}000 + 35{,}000(3.8887) + 15{,}000(0.4556) \\ &= \$2{,}938.50 > 0 \end{aligned}$$

$$\begin{aligned} PW_\Delta(16\%) &= -140{,}000 + 35{,}000\,(P/A,\, 16\%,\, 6) + 15{,}000\,(P/F,\, 16\%,\, 6) \\ &= -140{,}000 + 35{,}000(3.6847) + 15{,}000(0.4104) \\ &= -\$4{,}879.50 < 0 \end{aligned}$$

$$\begin{aligned} \Delta i^*_{C-A} &\approx 14\% + \frac{\$2{,}938.50}{\$2{,}938.50 - (-\$4{,}879.50)}\,(2\%) \\ &= 14\% + \frac{\$2{,}938.50}{\$7{,}818.00}\,(2\%) \\ &= 14.75\% \quad \text{(Excel: } 14.73\%) \end{aligned}$$

Since $\Delta i^*_{C-A} = 14.75\% \geq \text{MARR} = 12\%$, the additional \$140,000 investment is economically justified. **Package C replaces Package A as the current defender.**

Comparison 2: Challenger D vs. Defender C

$$\Delta P = 380{,}000 - 260{,}000 = \$120{,}000$$
$$\Delta A = 100{,}000 - 75{,}000 = \$25{,}000$$
$$\Delta S = 40{,}000 - 25{,}000 = \$15{,}000$$
$$\Delta A' = 25{,}000 + \frac{15{,}000}{6} = \$27{,}500/\text{year}$$
$$(P/A,\, i_0,\, 6) = \frac{\$120{,}000}{\$27{,}500} = 4.3636 \implies i_0 \approx 9.9\%$$

Evaluating incremental Present Worth $PW_\Delta(i)$ at 8% and 10%:

$$\begin{aligned} PW_\Delta(8\%) &= -120{,}000 + 25{,}000\,(P/A,\, 8\%,\, 6) + 15{,}000\,(P/F,\, 8\%,\, 6) \\ &= -120{,}000 + 25{,}000(4.6229) + 15{,}000(0.6302) \\ &= \$5{,}025.50 > 0 \end{aligned}$$

$$\begin{aligned} PW_\Delta(10\%) &= -120{,}000 + 25{,}000\,(P/A,\, 10\%,\, 6) + 15{,}000\,(P/F,\, 10\%,\, 6) \\ &= -120{,}000 + 25{,}000(4.3553) + 15{,}000(0.5645) \\ &= -\$2{,}650.00 < 0 \end{aligned}$$

$$\begin{aligned} \Delta i^*_{D-C} &\approx 8\% + \frac{\$5{,}025.50}{\$5{,}025.50 - (-\$2{,}650.00)}\,(2\%) \\ &= 8\% + \frac{\$5{,}025.50}{\$7{,}675.50}\,(2\%) \\ &= 9.31\% \quad \text{(Excel: } 9.29\%\text{)} \end{aligned}$$

Evaluating directly at MARR = 12%:

$$\begin{aligned} PW_\Delta(12\%) &= -120{,}000 + 25{,}000\,(P/A,\, 12\%,\, 6) + 15{,}000\,(P/F,\, 12\%,\, 6) \\ &= -120{,}000 + 25{,}000(4.1114) + 15{,}000(0.5066) \\ &= -\$9{,}616.00 < 0 \end{aligned}$$

Since $\Delta i^*_{D-C} = 9.31\% < 12\%$, the incremental \$120,000 outlay required for Package D fails the MARR hurdle. **Package C survives as the overall winner.**

Part 3. Select Package C (sorting robots). Although Package A boasts the highest individual i^* (25.26%), Package C represents the largest capital outlay whose incremental investments each earn at least the corporate MARR (12%).

Part 4. *The Junior Analyst Trap (Comparing D directly with A):*

$$\Delta P = 380{,}000 - 120{,}000 = \$260{,}000$$
$$\Delta A = 100{,}000 - 40{,}000 = \$60{,}000$$
$$\Delta S = 40{,}000 - 10{,}000 = \$30{,}000$$
$$\Delta A' = 60{,}000 + \frac{30{,}000}{6} = \$65{,}000/\text{year}$$
$$(P/A,\, i_0,\, 6) = \frac{\$260{,}000}{\$65{,}000} = 4.0000 \implies i_0 \approx 13.0\%$$

Evaluating at 12% and 14%:

$$\begin{aligned} PW_\Delta(12\%) &= -260{,}000 + 60{,}000\,(P/A,\, 12\%,\, 6) + 30{,}000\,(P/F,\, 12\%,\, 6) \\ &= -260{,}000 + 60{,}000(4.1114) + 30{,}000(0.5066) \\ &= \$1{,}882.00 > 0 \end{aligned}$$

$$\begin{aligned} PW_\Delta(14\%) &= -260{,}000 + 60{,}000\,(P/A,\, 14\%,\, 6) + 30{,}000\,(P/F,\, 14\%,\, 6) \\ &= -260{,}000 + 60{,}000(3.8887) + 30{,}000(0.4556) \\ &= -\$13{,}010.00 < 0 \end{aligned}$$

$$\begin{aligned} \Delta i^*_{D-A} &\approx 12\% + \frac{\$1{,}882.00}{\$1{,}882.00 - (-\$13{,}010.00)}\,(2\%) \\ &= 12\% + \frac{\$1{,}882.00}{\$14{,}892.00}\,(2\%) \\ &= 12.25\% \geq 12\% \quad \text{(Excel: } 12.24\%\text{)} \end{aligned}$$

Fatal Flaw in Logic: Once Package C beats Package A in pairwise comparison, Package A is strictly eliminated from further contention and can never serve as a baseline. The \$260,000 leap from A to D improperly bundles a highly lucrative increment ($A \to C$ earning 14.75%) with an unprofitable increment ($C \to D$ earning only $9.31\% < 12\%$). The surplus generated by the first step masks the value destruction of the second step, misleading an inexperienced analyst into selecting an economically inferior alternative.

Part 5. *Present Worth Verification at MARR = 12%:* Using $(P/A,\, 12\%,\, 6) = 4.1114$ and $(P/F,\, 12\%,\, 6) = 0.5066$:

$$PW_A(12\%) = -120{,}000 + 40{,}000(4.1114) + 10{,}000(0.5066) = \$49{,}522.00$$
$$PW_B(12\%) = -210{,}000 + 45{,}000(4.1114) + 15{,}000(0.5066) = -\$17{,}388.00$$
$$PW_C(12\%) = -260{,}000 + 75{,}000(4.1114) + 25{,}000(0.5066) = \mathbf{\$61{,}020.00}$$
$$PW_D(12\%) = -380{,}000 + 100{,}000(4.1114) + 40{,}000(0.5066) = \$51{,}404.00$$

Package C achieves the highest Present Worth ($61,020.00), perfectly corroborating the incremental rate of return conclusion.

Part C — Conceptual Multiple-Choice Questions

1. **Answer: (b).** i^* is the interest rate that brings the Present Worth of the project's net cash flows to zero ($PW(i^*) = 0$). Option (a) describes flat instalment financing, and (c) describes the corporate hurdle rate (MARR).

2. **Answer: (d).** In an unrecovered-balance table, interest charged in Year t is computed strictly on the unrecovered balance at the beginning of Year t. The ending balance decreases as cash receipts recover principal.

3. **Answer: (a).** Descartes' rule of signs establishes an *upper bound*: the maximum number of real roots where $i > -100\%$ equals the number of sign changes in the net cash flow series.

4. **Answer: (c).** Norstrom's criterion guarantees a unique positive rate of return ($i^* > 0\%$) if the cumulative cash flows S_t start negative ($S_0 < 0$) and change sign exactly once.

5. **Answer: (b).** When multiple roots occur with one negative and one positive rate, course guidelines specify using the positive rate (+18%) as the project's ROR. Since $18\% \geq 10\%$, the project is accepted.

6. **Answer: (d).** Setting $PW = -60{,}000 + 24{,}000\,(P/A,\, i^*,\, 3) = 0$ yields $(P/A,\, i^*,\, 3) = 2.5000$, which corresponds to $i^* = 9.70\%$. Distractor (a) is simple flat interest $(72k - 60k)/(3 \times 60k) = 6.67\%$, and (b) is annual cash flow divided by principal $(24k/60k = 40\%)$.

7. **Answer: (c).** Since the incremental rate of return $\Delta i^*_{B-A} = 12\%$ exceeds the MARR of 10%, the extra capital invested in alternative B is economically justified. Therefore, B is selected despite A having the higher individual ROR.

8. **Answer: (a).** Cost alternatives consist solely of disbursements (negative cash flows), so no positive internal rate of return exists. Consequently, the individual screening step is skipped, and the analyst proceeds directly to pairwise incremental comparisons.

9. **Answer: (d).** The Annual Worth of a single life cycle equals the Annual Worth over any integer number of cycles. Therefore, setting $AW_B(i) = AW_A(i)$ on single cycles satisfies the equal-service requirement without needing the Lowest Common Multiple (LCM).

10. **Answer: (b).** Excel's `RATE` function accepts arguments only for a uniform series

(`pmt`), an initial outlay (`pv`), and a final lump sum (`fv`). Cash flows with gradients or irregular timings require `IRR`.